

Intro to AI Engineering

Full module revision notes — from first prompt to secure backend.

CORE LOOP

1	2	3	4
Define behavior with prompts	Control quality, cost & speed	Build safe UX with error handling	Move calls to backend

KEY CONCEPTS

Prompt design • Clear constraints beat vague requests • Ask for structure when you need it • Prompt quality = output quality	Message roles • `system` — permanent behavior rules • `user` — the current request • `assistant` — model output history
Tokens • More tokens = higher cost + slower • Keep context lean and relevant • Constrain output length for speed	Output format • Free text is flexible but hard to parse • JSON improves machine readability • Structured outputs = most reliable

GIFT GENIE ARCHITECTURE

Frontend	Backend
1 Collect user prompt	1 Read userPrompt from req.body
2 POST to /api/gift with JSON body	2 Push user message into messages
3 Parse JSON response	3 Call model via OpenAI SDK
4 Convert markdown to HTML	4 Extract response content
5 Sanitize HTML with DOMPurify	5 Return { giftSuggestions }

COMMON MISTAKES

mistake	Mismatched field names — Use the same name on both sides — e.g. userPrompt everywhere
mistake	Rendering raw model text as HTML — Always parse markdown first, then sanitize with DOMPurify
mistake	Forgetting await response.json() — Parse async before reading any fields from the response
mistake	API key in client code — Move all model calls to the server — never expose keys in the browser
mistake	Poor output structure — Specify headings and a response template in the system prompt

REMEMBER JUST THESE 5 THINGS

1	Define behavior in the system message
2	Keep user input clean and specific
3	Enforce output format for UI reliability
4	Sanitize all rendered markdown
5	Move AI calls to backend before going live

REVISION DRILLS

Drill A 60-sec explain Talk through core concepts out loud: messages array, system vs user prompt, backend safety.	Drill B Rebuild from blank Recreate the frontend submit handler, backend route, and markdown-safe rendering from scratch.	Drill C Parameter test Run the same prompt at temperature 0, 0.7, and 1.0 — compare consistency and creativity.
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AI engineering = prompt quality + architecture quality + safety quality.